Badampeta Pranay Kumar

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Career Objective

A Computer Science and Engineering graduate with a strong interest in technology and problem-solving. Possesses good analytical thinking, time management, and teamwork skills, with the ability to adapt quickly and handle responsibilities effectively. Looking forward to learning, improving skills, and contributing to real-world projects in a professional environment.

Education

Woxsen University, Hyderabad 2021 – 2025
B.Tech in Computer Science and Engineering *65%*

Sri Gayatri Educational Institutions, Hyderabad 2019 – 2021
Intermediate (MPC) *89%*

Aruna High School, Hyderabad 2009 – 2019
Matriculation *88%*

Skills & Tools

- **Programming:** Python, Java, SQL
- **Frameworks:** PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO
- **Libraries:** NumPy, Pandas, Matplotlib
- **Tools:** Git, Jupyter Notebook, MySQL

Work Experience

Application Development Intern Mar 2024 – Jun 2024
Tricode Solutions, Hyderabad

- Developed responsive web applications using React.js.
- Built reusable components using React Hooks.
- Integrated REST APIs using Axios and Fetch.
- Managed state using `useState` and `useEffect`.
- Improved UI performance and fixed frontend bugs.
- Used Git for version control and collaboration.

Social Internship Jun 2023 – Aug 2023
Vishwa Bharathi High School, Zaheerabad, India

- Inspected and analyzed internet usage on Android smartphones.
- Collected data on app usage, screen time, and data consumption.
- Prepared reports and presented insights to improve digital awareness among students and staff.
- Promoted safe and responsible internet usage practices.

Projects

Stock Price Prediction Using Machine Learning

- Built prediction models using Linear Regression and LSTM to forecast stock prices.
- Performed data preprocessing, feature scaling, and time-series analysis.
- Evaluated model performance using RMSE and MAE.
- Visualized trends and prediction results using Matplotlib to interpret model behavior and insights.

Credit Card Fraud Detection Using Machine Learning

- Developed fraud detection models using Logistic Regression, Random Forest, and XGBoost.
- Handled imbalanced transaction datasets using SMOTE.
- Evaluated performance using precision, recall, F1-score, and confusion matrix.

Digital Signature Using Hashing

- Designed a digital signature system using SHA-256 hashing and RSA encryption.
- Ensured data integrity and authentication during secure communication.
- Implemented secure key generation, signing, and verification mechanisms.

Web-based Interface for Detecting and Classifying Solar PV Panel Defects

- Developed an IoT and ML-based solar PV monitoring system using CNN models.
- Implemented fault detection using Random Forest and SVM.
- Analyzed real-time sensor data to improve performance and maintenance decisions.
- Research work published in an IEEE conference/journal.

Publications

Web-based Interface for Detecting and Classifying Solar PV Panel Defects Using CNN Model

2025 IEEE 9th International Conference on Information and Communication Technology (CICT)

- Developed a CNN-based solar PV defect detection system.
- Deployed the model through a web-based interface for real-time monitoring and analysis.

Certifications

- NASA International Space Apps Challenge 2024 – Galactic Problem Solver *Oct 2024*
- Programming Foundations with JavaScript, HTML, and CSS *Jun 2022*
- Data Analysis with Python *Dec 2022*
- Data Visualization with Python *Mar 2023*
- AI for Everyone *Apr 2023*

Achievements

- AGNIRVA 2025 Space Internship Program – National-level internship focused on satellite data, space analytics, and AI-driven applications.
- NASA International Space Apps Challenge 2024 – Galactic Problem Solver.